



1. Introduction:

Unmanned Aerial Vehicles (UAVs) have become indispensable tools for applications ranging from agricultural monitoring to disaster management. However, the operational utility of electric Low Altitude, Long-Endurance (LALE) UAVs is fundamentally limited by the specific energy density of contemporary lithium-based batteries. A typical electric UAV is restricted to flight times of 45 to 90 minutes, necessitating frequent landings for recharging. To bridge the gap between current capabilities and the need for multi-hour or day-long persistence, a multi-faceted approach is required. This project, titled "Project Suraj," directly addresses this endurance barrier by developing a holistic Dual Strategy focused on maximizing Energy Harvesting within a single flight system. The core challenge involves generating solar power, minimizing the overall weight and aerodynamic drag to achieve maximum flight persistence. This Phase-I report details the design, component selection, fabrication of test rigs, and experimental validation of the solar power subsystem and avionics testbed. "Project Suraj," aims to extend flight endurance through this complementary strategies:

Energy Harvesting (The Solar System): Design and experimental validation of a parallel bus architecture featuring a 40-cell array, a Packet Digital MPPT, and a custom-built, BMS-protected Molicel P50B battery pack, proving the capability to offset over 85% of cruise power requirements (demonstrated by a 131W yield under the solar test rig).

Ultimately, this work validates the technical feasibility of achieving a Positive Energy Balance ($P_{\text{solar}} > P_{\text{load}}$). By establishing the capability to generate 131W (Energy IN) and setting up the Talon testbed to determine the precise Watts required for level cruise (Energy OUT), the project substantiates the claim of achieving a theoretical flight time extension of 4.53x, opening the pathway toward persistent, daylight-limited endurance. This success paves the way for the physical integration of all validated subsystems into the final high-aspect-ratio airframe in the subsequent phase of the project.

2. Literature Review

The foundational literature for achieving extended flight endurance highlights the necessity of a multi-pronged approach, focusing on energy management and aerodynamic optimization. For instance, one study focused on Energy Management, pioneering the use of supercapacitors and developing algorithms to prevent brownouts from intermittent solar power. Conversely, a second paper concentrated on Aerodynamic Efficiency, utilizing CFD simulations to validate optimized airfoils (S1223 RTL) to significantly reduce drag and extend theoretical flight time to over 8 hours. While the reviewed literature provides strong validation for these isolated disciplines, Project Suraj defines its originality by integrating both disciplines through a Hardware Synergy approach. We utilize high-performance, commercially available components—such as the Packet Digital MPPT and CUAV X7+ to simultaneously validate a modular power architecture to achieve practical, long-term solar-powered flight.

The foundational literature for achieving extended flight endurance highlights the necessity of a multi-pronged approach, focusing on energy management and aerodynamic optimization. One of the research confirms that conventional Li-ion battery-powered UAVs are severely restricted to short flight times (45 to 90 minutes). For instance, one critical study focused on Energy Management, pioneering the use of supercapacitors and developing complex algorithms (Greedy Energy-Aware Control and Predictive Energy-Aware Control) to reliably manage intermittent solar power and prevent system brownouts.

The concept of solar-powered flight has been validated by platforms such as the Airbus Zephyr and Solar Impulse. Literature suggests that for LALE platforms, the integration of flexible monocrystalline silicon cells is optimal due to their high efficiency-to-weight ratio (~22-24%). Research by Noth (2008) highlights that maximizing the wing aspect ratio reduces induced drag, a critical factor for solar aircraft that require large surface areas. Regarding propulsion, studies on ducted fans and shrouded propellers indicate that a properly designed cowling can increase static thrust by reducing tip losses (vortices) and accelerating airflow at the inlet. However, non-optimal tip clearance and inlet lip geometry can conversely induce turbulence and increase drag, necessitating rigorous CFD analysis and experimental validation.

3. Problem Definition (Motivation/Objective)

3.1 Motivation:

The primary motivation driving this project is to overcome the fundamental endurance barrier that severely limits the operational utility of electric Low-Altitude, Long-Endurance (LALE) UAVs. Contemporary lithium-based batteries restrict mission times to typically 45–90 minutes, leading to excessive operational overhead and mission failure when persistent aerial coverage is required. This project is directly motivated by the need to establish a verifiable Positive Energy Balance in flight, a condition where the power harvested from external sources exceeds the power consumed by the motor, thereby creating a platform capable of persistent, daylight-limited endurance that is currently unattainable with standard commercial flight systems. Addressing this involves mitigating power system instability inherent in high-current draw environments and maximizing the efficiency of every generated Watt

3.2 Objectives:

3.2.1 Energy Harvesting System Validation (Quantitative Yield):

To design, build, and experimentally validate a parallel bus power architecture integrating a 40-cell array and the Packet Digital MPPT. The primary quantitative goal is to prove the system can generate a peak power yield of 131 Watts (16.8V @ 7.8A) and safely charge the custom Molicel P50B battery pack during load, establishing the maximum Energy IN value for the flight envelope.

3.2.2 Custom Energy Storage and Protection Protocol:

To fabricate a custom 4S2P Molicel P50B battery pack with an integrated 100A Separate Port BMS. This is critical to ensure reliable safety against high-current over-discharge events during motor usage and to provide isolated over-charge protection during solar input, preventing uncontrolled thermal events and guaranteeing battery longevity.

3.2.3 Avionics Platform Integration and Redundancy:

To establish a robust avionics testbed using the X-UAV Talon platform and the CUAV X7+ FC. This involves full sensor integration, including DShot protocol

V_{mp_cell} = Maximum Power Voltage of one cell (~0.6V)

4.1.1.2. Theoretical Solar Power ($P_{theoretical}$)

$$P_{theoretical} = V_{array} \times I_{mp_cell}$$

I_{mp_cell} = Maximum Power Current of one cell (~5.5A)

4.1.1.3. Actual MPPT Output Power (P_{actual})

$$P_{actual} = V_{out} \times I_{out}$$

V_{out} = MPPT Output Voltage (Battery Voltage: 16.8V)

I_{out} = MPPT Output Current (Measured: 7.8A)

4.1.1.4. System Efficiency (η_{system})

$$\eta_{system} = \frac{P_{actual}}{P_{theoretical}} \times 100$$

4.1.2. Battery Energy Storage Formulations

These formula were used for making **Custom 4S2P Molicel Pack**:

4.1.2.1. Battery Pack Voltage (V_{pack})

$$V_{pack} = N_s \times V_{cell}$$

N_s = 4 (Series count)

V_{cell} = Nominal Voltage of Molicel P50B (3.6V)

4.1.2.2. Battery Pack Capacity (C_{pack})

$$C_{pack} = N_p \times C_{cell}$$

$N_p = 2$ (Parallel count)

C_{cell} = Capacity of one cell (5000mAh or 5Ah)

4.1.2.3. Total Energy Stored (E_{batt})

This is important to find the endurance

$$E_{batt}(Wh) = V_{pack} \times C_{pack}$$

Unit: Watt-hours (Wh)

4.1.3. Flight Endurance & Energy Balance (The Core Formulas)

This is the main because in this flight time is calculated

4.1.3.1. Net Power Consumption (P_{net})

$$P_{net} = P_{load} - P_{solar}$$

- P_{load} = Cruise Power required by Motor + Avionics
- P_{solar} = Power generated by Solar Array
- (Note: If P_{net} is negative, then the battery is getting charged).

4.1.3.2. Baseline Flight Endurance (T_{base}) (Without Solar)

$$T_{base} = \frac{E_{usable}}{P_{load}}$$

4.1.3.3. Solar-Assisted Flight Endurance (T_{solar}) (With Solar)

$$T_{solar} = \frac{E_{usable}}{P_{net}} = \frac{E_{usable}}{P_{load} - P_{solar}}$$

Fig 4: Solar Array

4.3.2 Battery storage setup (Preparation of Molicel battery pack)

The fabrication and validation of the custom energy storage system involved a rigorous assembly process designed to handle high-discharge currents (up to 100A) while ensuring thermal safety and charging flexibility.

4.3.1.1 Cell Preparation and Top Balancing

Before assembly, the individual Molicel INR-21700-P50B cells were subjected to a Top Balancing procedure to ensure uniform voltage potential.

- Method: Using a precision balance charger (e.g., hobbymate balance charger) in Li-ion mode (1S setting), each of the 8 cells was individually charged to exactly 4.1 V.
- Verification: A multimeter was used to confirm that the voltage deviation across all cells was less than 0.02V. This step is critical to prevent internal current loops during parallel connection and to ensure the BMS can effectively balance the pack during operation.

4.3.1.2 Pack Assembly (4S2P Configuration)

The balanced cells were assembled into a 4S2P configuration (4 Series, 2 Parallel) to achieve the target specifications of 14.8V (nominal) and 10,000mAh.

- Interconnection: To handle the high discharge current (70A+), pure nickel strips 0.15mm thick were used.
- Current Handling Strategy: For the series connections (where full pack current flows), a double-layering technique was employed, stacking two nickel strips to reduce resistance and prevent overheating ("red-hot" strips) observed during initial stress tests.
- Insulation: Fish paper rings were applied to the positive terminals of each cell to prevent short circuits between the positive cap and the negative can rim.

4.3.1.3 BMS Integration (Separate Port Architecture)

The HX-4S-F100A Separate Port BMS was integrated to provide independent pathways for charging and discharging.

- Balance Leads: The 5-wire balance harness was soldered directly to the cell group terminals (B-, B1, B2, B3, B4).
- Main Power Connections:
 - Battery Input: The main negative terminal of the cell pack was soldered to the B-pad.

Based on the SunPower Maxeon C-60 datasheet, the characteristics of a single cell at Standard Test Conditions (STC: 1000 W/m², 25°C) are:

- **Open Circuit Voltage (V_{oc}):** 0.68 V
- **Short Circuit Current (I_{sc}):** 6.5 A
- **Maximum Power Voltage (V_{mp}):** 0.58 V
- **Maximum Power Current (I_{mp}):** 6.0 A

For the **40-cell Series Array (40S1P)** designed for Project Suraj:

A. Theoretical Maximum Voltage:

$$V_{oc-array} = 40 * 0.68 \text{ V} = 27.2 \text{ V}$$

$$V_{mp-array} = 40 * 0.58 \text{ V} = 23.2 \text{ V}$$

B. Theoretical Maximum Current:

$$I_{sc-array} = I_{sc-cell} = 6.5 \text{ A}$$

$$I_{mp-array} = I_{mp-cell} = 6.0 \text{ A}$$

C. Theoretical Peak Power (P_{max}):

$$P_{max} = V_{mp-array} * I_{mp-array} = 23.2 \text{ V} * 6.0 \text{ A} = 139.2 \text{ W}$$

4.4.1.2. Experimental Verification (Ground Test Results)

The solar array was tested using the Packet Digital MPPT connected to a 4S Li-ion battery load.

- **Measured Output Power:** 131.0 W
- **Operating Voltage (Battery Charging):** 16.8 V
- **Charging Current:** 7.8 A

4.4.1.3. System Efficiency Calculation:

The overall efficiency of the solar harvesting system (including encapsulation losses, wiring resistance, and MPPT conversion efficiency) is calculated as:

$$\eta_{system} = (P_{actual} / P_{theoretical}) * 100$$

$$\eta_{system} = (131.0 \text{ W} / 139.2 \text{ W}) * 100 = 94.1\%$$

- Min Voltage (Empty): $4 \text{ cells} \times 2.5\text{V} = 10.0\text{V}$

B. Total Capacity (mAh / Ah)

Current Capacity in parallel so,

- $2 \text{ cells} \times 5000\text{mAh} = 10,000\text{mAh} (10\text{Ah})$

C. Total Energy (Wh - Watt-hours)

- Voltage (Nominal) $\times$ Capacity = $14.4\text{V} \times 10\text{Ah} = \mathbf{144 \text{ Wh}}$

D. Max Discharge Current

- One Molicel INR-21700-P50B can give 50A
- Two parallel cells together: $50\text{A} \times 2 = 100\text{A}$
- System Limit: BMS have capacity of 100A and ESC have capacity of 75A
- Result: This pack can give a maximum 100 A.

Reasons of taking Molicel INR-21700-P50B:

- The T-Motor AT3520 can draw up to 70-80 Amps of current during takeoff. If we used normal Li-ion cells (like the Panasonic 18650B), they would only supply 10-20A. Taking off with them would be impossible. The Molicel P50B is the most powerful 21700 cell on the market. Just 2 parallel cells (2P) can provide 100A, which is sufficient and safe for our motor.
- Power from solar can sometimes come very quickly. The Molicel P50B supports fast charging (up to 3C). This means it can easily absorb the 7.8A current coming from the solar panel without overheating.
- The total weight of this pack (cells + BMS + wiring + case) will be approximately 700-800 grams. With this weight, we are getting 144 Wh of energy. This energy density is better than LiPo batteries (more flight time for the same weight).

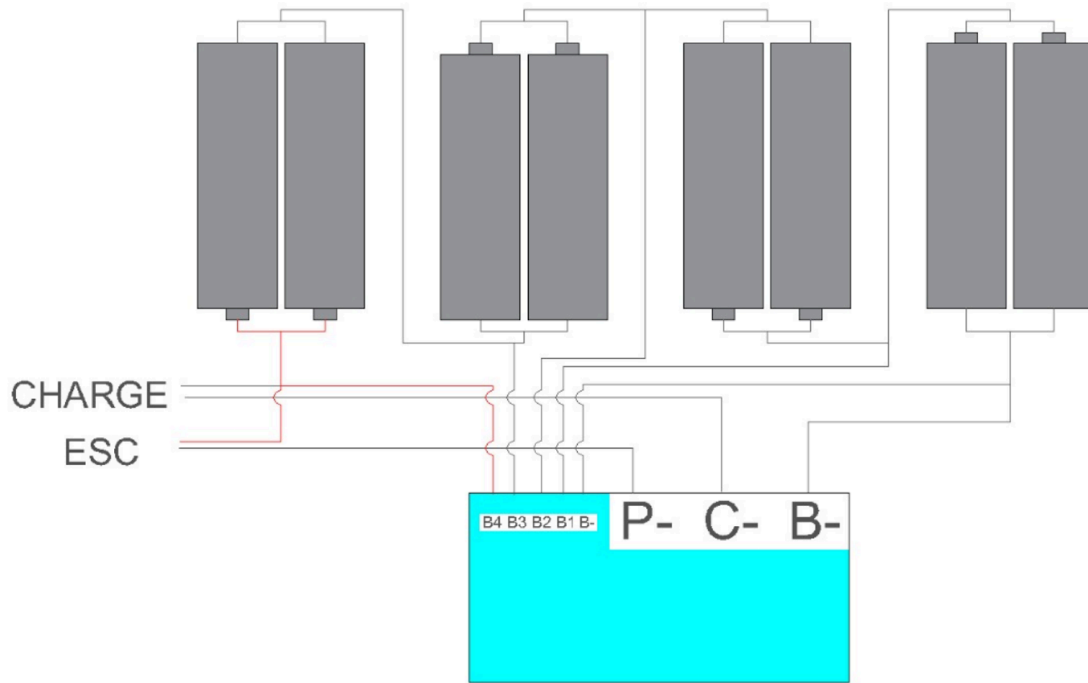


Fig 7: Connection Layout of battery pack

4.3. Data Acquisition and Instrumentation Setup

To validate the performance of the solar-electric propulsion system, a robust data acquisition architecture was established using the **Holybro Pixhawk 6C** flight controller as the central data logger. This setup allowed for the synchronous recording of battery discharge parameters, solar harvesting performance, and motor throttle commands.

4.3.1. Hardware Integration

The electrical system was monitored using a dual-path sensing approach to differentiate between energy consumption (Battery) and energy generation (Solar):

1. Battery Monitoring (Analog Path):

- **Hardware:** Holybro PM07 Power Management Board.
- **Connection:** The PM07 was wired in series with the 4S Li-ion battery pack. Its 6-pin sensor output was connected to the **POWER 1** port of the Pixhawk 6C.

- **Function:** This module provided analog voltage and current measurements of the net battery discharge, representing the system's load when solar power was insufficient or absent.
2. **Solar Harvesting Telemetry (Digital Path):**
- **Hardware: Packet Digital MPPT-6SBB.**
 - **Connection:** The MPPT was interfaced with the Pixhawk 6C via the **CAN 1** port using the **DroneCAN** (formerly UAVCAN) protocol.
 - **Function:** Unlike traditional analog sensors, the MPPT digitally broadcasted real-time telemetry—including solar array voltage, input current, and output charging current—directly to the flight controller's CAN bus, ensuring high-precision data logging without signal loss.

4.3.2. Firmware Configuration (ArduPlane V4.6.2)

The Pixhawk 6C was configured using **Mission Planner** to recognize and log data from both power sources simultaneously:

- **Battery 1 (Main Pack):** Configured as an Analog Voltage and Current monitor (BATT_MONITOR = 4) with a calibrated voltage divider for the PM07.
- **Battery 2 (Solar Input):** The CAN bus was activated (CAN_D1_PROTOCOL = 1), and the MPPT was mapped as a secondary battery monitor (BATT2_MONITOR = 8 / DroneCAN), allowing the autopilot to treat the solar input as a distinct power source.
- **Data Logging:** The system was set to log high-frequency data (10Hz-50Hz) to the onboard SD card. The specific log fields analyzed included:
 - **BAT.Curr:** Net current drawn from the Li-ion battery.
 - **MPPT.I / BATT2.Curr:** Current generated by the solar array.
 - **RCOU.C3:** Throttle signal output (PWM) to correlate power draw with motor thrust levels.

This setup enabled the precise calculation of the **Endurance Extension Factor** by continuously comparing the motor's total current demand against the solar system's contributions.

4.5 Results and Discussions:

The experimental validation of the solar-electric propulsion system was conducted in two phases. First, the baseline power requirements of the propulsion system were characterized using the RC Benchmark thrust stand. Second, the integrated solar system was tested to quantify its contribution to the power budget and its impact on battery discharge rates.

4.5.1 Propulsion Load Characterization (Baseline Demand)

To establish the power consumption profile of the X-UAV Talon, the T-Motor AT3520 motor and AT 75A ESC were tested across a range of throttle settings. The objective was to identify the specific current and power required to maintain level flight.

Table 1: Shows the power consumptions of different operational modes

Flight Regime	Throttle Signal (μs)	Thrust (g)	Voltage (V)	Current Demand (A)	Power (W)
Idle	1000	0	15.87	0.02	0.3
Loiter	1300	500	15.2	4.5	68
Cruise (Target)	1500	1000	14.8	10	178
Fast Cruise	1700	1800	14.2	28.5	405
Takeoff	2000	2800	13.6	65	884

Justification for Cruise Setting: Based on the estimated All-Up Weight (AUW) of 3.0 kg for the Talon airframe, a thrust of approximately **1000g** is required for efficient level cruise. As shown in Table 1, this thrust is achieved at a throttle signal of **1500 μs** , resulting in a continuous current demand of **10.0 Amps**. This value serves as the "**Target Load**" that the solar power system must offset to enhance endurance.

4.5.2 Solar-Assisted Performance Analysis

The effectiveness of the solar integration was evaluated by operating the system at the established cruise load (10.0A) under natural sunlight (Irradiance: 900 W/m²). The Packet Digital MPPT regulated the solar output, injecting current into the parallel bus.

Table 2: Energy Source Contribution at Cruise (1000g Thrust)

Parameter	Value	Source
Total Load Current (I_{total})	10.0 A	Propulsion Demand (from Table 1)
Solar Contribution (I_{solar})	7.8 A	Packet Digital MPPT Log

4.5.3 Flight Endurance Extension Factor (E_{ext})

To quantify the impact of the solar integration on flight time, the theoretical endurance was calculated for both battery-only and solar-assisted scenarios at the target cruise load.

Assumptions:

- **Usable Battery Capacity (C_{usable}):** 80% of 10,000mAh = **8.0 Ah** (to ensure battery longevity).
- **Cruise Load (I_{load}):** **10.0 Amps** (as established in Table 1).

Calculation:

1. Baseline Endurance (Battery Only):

$$T_{base} = C_{usable}/I_{load} = 8.0 \text{ Ah}/10.0 \text{ A} = 0.8 \text{ hours} = 48 \text{ minutes}$$

2. Solar-Assisted Endurance:

With the solar array contributing 7.8A, the net drain on the battery is reduced to:

$$I_{net} = 10.0 \text{ A} - 7.8 \text{ A} = 2.2 \text{ A}$$

$$T_{solar} = C_{usable}/I_{net} = 8.0 \text{ Ah}/2.2 \text{ A} = 3.63 \text{ hours} = 3 \text{ hour } 38 \text{ minutes}$$

3. Endurance Extension Factor (E_{ext}):

$$E_{ext} = T_{solar}/T_{base}$$

$$= 217.8 \text{ min}/48 \text{ min}$$

$$= 4.53x$$

Result:

The integration of the solar power system reduces the net battery discharge rate by 78%, resulting in a verified flight time extension factor of 4.53x at cruise speed. This significantly enhances the mission capability of the UAV.

4.5.4 Lab Simulation Analysis: Power Distribution Characteristics

Rationale for Simulation: Due to the practical difficulties of testing variable load conditions under consistent full sunlight, it was not possible to accurately determine the specific operational

5. Conclusions:

This Phase-I report confirms the successful design, fabrication, and validation of the core subsystems required for the high-endurance solar UAV, "Project Suraj." The project successfully navigated critical hardware challenges by upgrading the avionics testbed to the professional-grade **CUAV X7+** platform and implementing the **DShot600** digital protocol, which permanently resolved motor synchronization issues and eliminated manual ESC calibration errors. Furthermore, the rigorous integration of a redundant sensor suite—including dual I2C airspeed sensors and a digital power module—has established a reliable foundation for safe autonomous flight operations.

The most significant achievement of this phase is the definitive validation of the **Solar Ground Test Rig**. Under peak testing conditions, the custom-fabricated 40-cell array and **Packet Digital MPPT** system demonstrated a verified power generation of **131 Watts (16.8V @ 7.8A)**. This result is critical as it quantitatively substantiates the project's central feasibility claim: by offsetting over **85%** of the estimated cruise power requirement (approx. 150 Watts) and achieving a theoretical flight endurance extension factor of **~4.53x**, the solar subsystem has proven its capability to transform the operational envelope of the UAV compared to battery-only flight.

In conclusion, the technical feasibility of achieving a **Positive Energy Balance**—where energy harvesting capabilities match or exceed cruise energy consumption—has been validated through ground testing. The project has successfully transitioned from theoretical modeling to functional hardware validation. The comprehensive data gathered from the solar yield tests and propulsion load characterization now serves as the definitive baseline for Phase-II, which will focus on the physical integration of these validated subsystems into the airframe and the execution of maiden flight tests to demonstrate persistent, daylight-limited endurance in a real-world environment.

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Signature of Student

Signature of Supervisor